Restricted-Product Epsilon Duality

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Summary

Pass 71 formulates the all-prime signed duality statement for the Loeb-Rosser class without treating $\widehat{\mathbb{Z}}/\mathbb{Z}$ as an ordinary Hausdorff locally compact quotient.

The all-prime object is

$$ ¥epsilon_{¥mathbb P}

{¥epsilon_S}_{S¥subset¥mathbb P,¥ |S|<¥infty} $$

together with the derived pro-cokernel

$$\varprojlim^1 (N_n\mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}.$$

For finite S, each shadow remains the Pass-70 extension

$$0 \to \mathbb{Z}^S/\Delta\mathbb{Z} \to \widehat{\mathbb{Z}}_S/\mathbb{Z} \to \prod_{p\in S}(\mathbb{Z}_p/\mathbb{Z}) \to 0.$$

Support-Preserving Duality

The ordinary dual of an infinite product only sees finite-support continuous characters. Therefore the all-prime duality cannot be stated as a bare product duality.

Instead, it must use restricted products with conductor and lattice data:

$$\prod_p{}' (A_p, L_p).$$

At conductor k, the local finite window is

$$p^{-k}\mathbb{Z}_p/p^k\mathbb{Z}_p,$$

and the integral lattice

$$\mathbb{Z}_p/p^k\mathbb{Z}_p$$

is self-annihilating under the normalized conductor pairing.

Signed Law

Choose a base prime $p_0 \in S$. The finite boundary is

$$d_S(x) = (x_p - x_{p_0})_{p\neq p_0}.$$

The all-prime statement

$$ D_{¥mathrm{res}}(¥epsilon_{¥mathbb P})

-¥epsilon_{¥mathbb P}^{¥vee} $$

means that every finite prime/conductor shadow satisfies

$$D(d_S) = -d_S^T,$$

duality squared returns d_S, and all finite-prime restriction squares commute.

This is a pro-restricted finite-shadow theorem. It is not yet a full proof in a selected LCA-sheaf, condensed, solid, or hybrid exact category.

Verification

The checker `code/scripts/check-pass71.py` produces `artifacts/reports/pass71-restricted-product-epsilon-duality-check.json`.

It verifies that:

- finite boundary matrices commute with prefix restriction through six primes;
- signed transposes commute with the same restrictions;
- conductor windows for $p = 2, 3, 5, 7$ and $k = 1, 2$ have self-annihilating integral lattices;
- finite-prefix support counts separate restricted-product product profiles from bounded finite-support dual profiles.

The remaining problem is categorical: construct the ambient exact category and prove that this finite-shadow law is an actual all-prime duality theorem there.